

VARUN KUMAR S R

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PROFILE SUMMARY

Electronics and Communication Engineering student with strong interests in Embedded Systems, IoT, Artificial Intelligence, and Secure Communication Systems. Experienced in developing hardware-software integrated solutions using ESP32, Arduino, Python, Firebase, and Supabase. Passionate about building innovative technology products and solving real-world engineering problems through practical project development.

EDUCATION

Bachelor of Engineering (ECE) Maharaja Institute of Technology Mysore Current CGPA: 6.5	2027
Pre-University Course (PCMB) JSS Pre-University College, Mysore Percentage: 65.67%	2022
SSLC Percentage: 65.44%	2020

SKILLS

- **Programming:** C, Python, Arduino Programming
- **Embedded Systems:** ESP32, ESP8266, Arduino, Sensor Interfacing
- **Communication Protocols:** UART, SPI, I2C, ESP-NOW, OneWire
- **PCB Design:** EasyEDA, KiCad (Schematic to PCB workflow)
- **Tools:** Git, GitHub, VS Code, Firebase, Supabase
- **Domains:** IoT, Embedded Systems, Robotics, Cybersecurity, Post-Quantum Cryptography

PROJECTS

Quantum Secure Communication (QSC App) <i>React Native · Node.js · Supabase · Post-Quantum Cryptography</i>	GitHub (Private)
• Developed a cross-platform secure messaging application integrating Post-Quantum Cryptography (PQC) for future-resilient IoT communication. • Implemented hybrid encryption pipeline using CRYSTALS-Kyber (key exchange), Dilithium (digital signatures), and AES-256-GCM (authenticated encryption). • Integrated ANU Quantum Random Number Generator (QRNG) API for high-entropy cryptographic seed generation. • Designed real-time encrypted message synchronization using Supabase with a secure Node.js/Express backend. • Implemented ephemeral message expiration logic to enhance forward secrecy and reduce long-term data exposure.	
RF Scanner Application <i>Python · ESP32 · nRF24L01 · Signal Processing</i>	GitHub (Public)
• Built a wireless RF signal monitoring and analysis tool targeting the 2.4 GHz frequency band. • Implemented real-time signal detection, channel scanning, and RSSI-based visualization. • Designed for use in RF interference detection and localization applications.	
DSP Gesture Controller <i>Python · MediaPipe · IIR/FIR Filtering · OpenCV</i>	GitHub (Public)
• Developed a hand gesture recognition system using Python and MediaPipe for gesture-based computer control. • Applied IIR and FIR digital filters for noise reduction and gesture signal stabilization. • Implemented per-finger curl detection, palm orientation tracking, and an 8-frame stability check with 2.5-second action lock. • Packaged as a one-liner PowerShell installer with auto-update support via GitHub.	
Smart Overvoltage Protection System <i>Arduino Nano · MOSFET Switching · Hysteresis Control · Embedded C</i>	GitHub (Private)
• Designed a motor protection system using real-time voltage monitoring and MOSFET-based load control. • Implemented dual-threshold hysteresis logic to prevent unstable switching near overvoltage conditions. • Developed automatic cutoff and recovery mechanism based on configurable voltage limits.	

- Integrated 16x2 I2C LCD for live voltage display and applied ADC-based sensing for safe high-voltage measurement.

WiFi Scanner

GitHub (Private)

ESP32 · Python · Network Analysis

- Developed a WiFi network scanning and monitoring system to detect nearby access points.
- Displayed SSID, BSSID, channel, and RSSI signal strength information in real time.
- Designed for network security auditing and wireless environment analysis.

Smart Biometric Multi-Lock IoT System

GitHub (Private)

Arduino Mega · Fingerprint Sensor · EEPROM · Multi-Relay Control

- Designed a multi-user biometric smart locking system using Arduino Mega 2560 with two-factor authentication (Fingerprint + PIN).
- Implemented role-based access control (Admin/User) with EEPROM-based credential storage.
- Developed multi-zone architecture supporting 3 independent solenoid locks with UART and I2C communication interfaces.
- Designed relay driver and 12V solenoid power isolation circuit using EasyEDA; optimized OLED rendering to eliminate display flicker.

HACKATHONS & COMPETITIONS

- **Smart India Hackathon (SIH) – Hardware Category**

Proposed an intelligent pesticide spraying drone system using AI-based plant disease detection and RTK-GPS precision targeting. Focused on system architecture design, feasibility planning, and agricultural automation concepts.

- **Bahubali Engineering College Hackathon**

Participated in a hardware-focused hackathon competition, applying embedded systems and IoT knowledge to solve real-world engineering challenges.

ACHIEVEMENTS

- Developed 7+ engineering projects published on GitHub (VK-Innovate).
- Active contributor to Embedded Systems, IoT, and Cybersecurity projects.

CERTIFICATIONS

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| • Pressure, Force, Motion and Humidity Sensors – Coursera | View Certificate |
| • Probability and Statistics – Coursera | View Certificate |
| • Computer Networks and Network Security – Coursera | View Certificate |